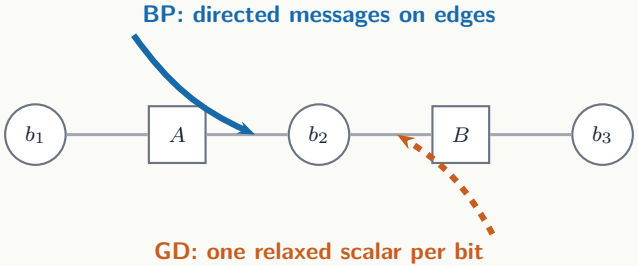
Same Tanner Graph, Different State

Why Tanner-graph traversal is not a sufficient mechanism



The graph specifies who is connected. It does not specify what is represented, retained, or used to calculate the next state.

Research question: what mechanism, or combination of mechanisms, differentiates BP from projected gradient descent?

A performance ordering cannot identify the mechanism



A native BP–GD comparison changes several boxes at once. It can show that the methods behave differently, but it cannot say which box caused the difference.

Invalid shortcut

one native method ranking

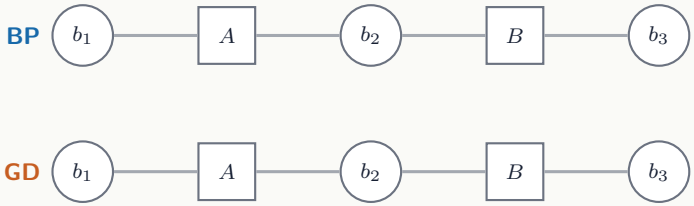
⇓

graph traversal is the cause

We need a controlled change, not an explanation inferred from the final ranking.

Evidence: RQ1-ENDPOINT-001 experiment card; mechanism report, lines 7–28

Both methods use the same Tanner-graph edges



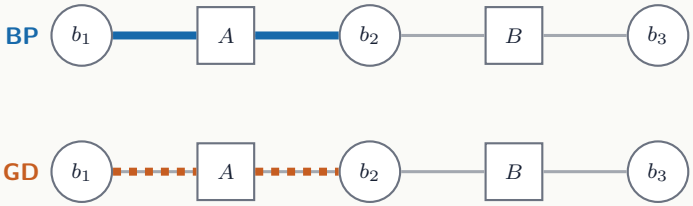
BP forms variable-to-check and check-to-variable messages on directed edges.

Projected GD uses neighbouring scalar values when evaluating partial derivatives of its objective function.

The Tanner graph organizes both calculations; GD does not need a separate "gradient graph."

Evidence: mailboxes_and_arrows.html; mechanism report, lines 313–322

Both methods use the same Tanner-graph edges



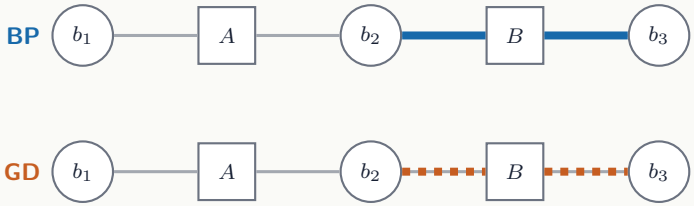
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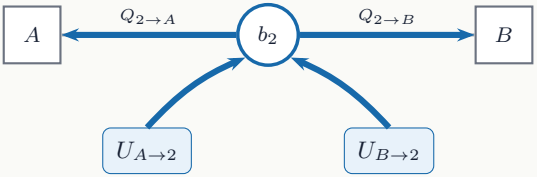
BP forms variable-to-check and check-to-variable messages on directed edges.

Projected GD uses neighbouring scalar values when evaluating partial derivatives of its objective function.

The Tanner graph organizes both calculations; GD does not need a separate "gradient graph."

Evidence: mailboxes_and_arrows.html; mechanism report, lines 313–322

BP: the recipient determines the message



$$Q_{2\to A} = L_2 + U_{B\to 2},$$
$$Q_{2\to B} = L_2 + U_{A\to 2}.$$

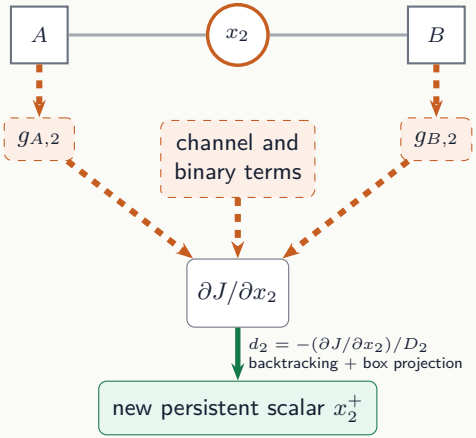
Sign convention: $L_i = \log \frac{P(b_i=1|y_i)}{P(b_i=0|y_i)}$; positive L_i favors bit one.

- The message to A omits the preceding message from A.
- The message to B omits the preceding message from B.
- One bit can therefore send different values to different checks.

BP augments each Tanner-graph edge with separately addressed directed message coordinates.

Evidence: RQ1-STATE-002; mechanism report, lines 44–65

Projected GD: one scalar state per bit



Native coordinate: $\ell = -\mathbf{L}$ and $x_i > 0$ represents bit zero; $D_i = (2\mu_p \deg(i) + 8\mu_b)/n$.

$$J(\mathbf{x}) = -\frac{\ell^{\mathsf{T}}\mathbf{x}}{2n} + \frac{\mu_p}{n}\sum_c \left(1 - \prod_{j\in\mathcal{N}(c)} x_j\right)^2 + \frac{\mu_b}{n}\sum_i (1-x_i^2)^2,$$

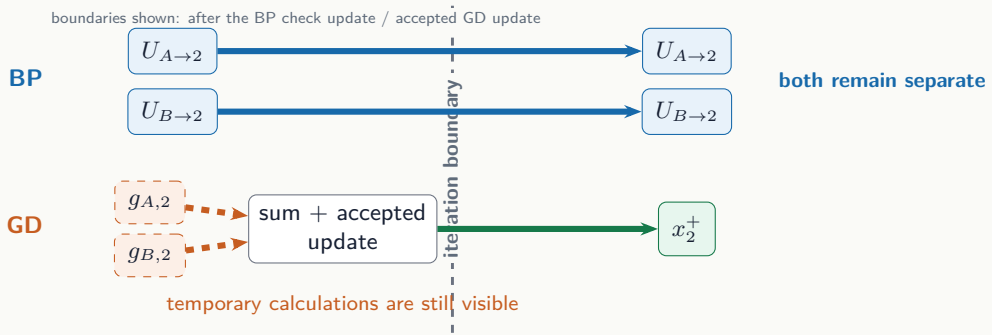
$$\frac{\partial J}{\partial x_2} = g_{A,2} + g_{B,2} + g_{\mathrm{ch},2} + g_{\mathrm{bin},2}.$$

- All terms use the same frozen $\mathbf{x}$ and are combined for this update.
- Only the accepted projected x_2^+ enters the next update.

The quantities $g_{A,2}$ and $g_{B,2}$ contribute to a partial derivative of the objective function. They are not BP messages, and their summation is the declared operation for this scalar objective.

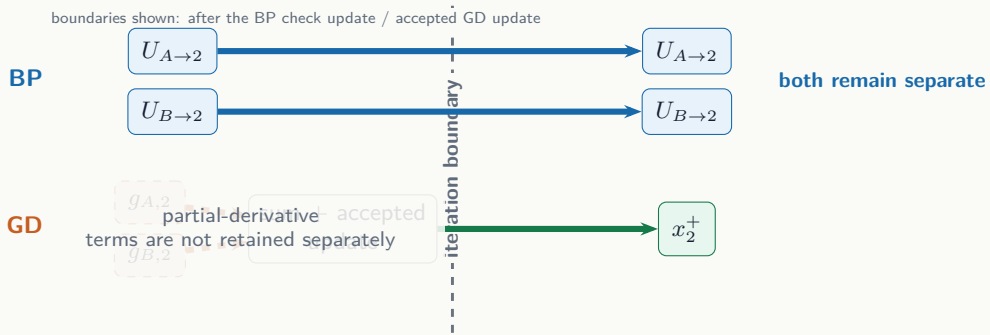
Evidence: RQ1-ENDPOINT-PG-001 method description and validated native trace

The decisive visual: what crosses the iteration boundary?



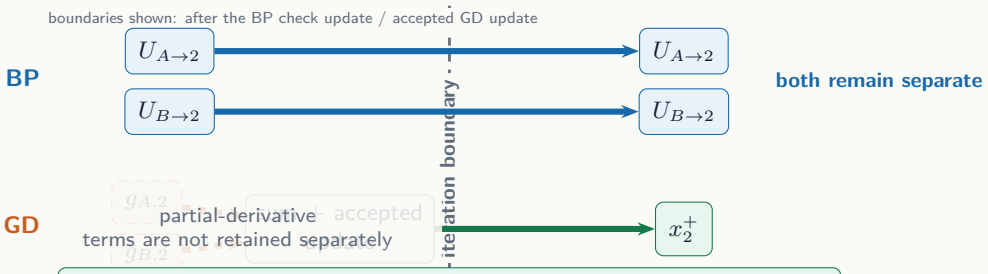
Evidence: mailboxes_and_arrows.html, beats 19–20; RQ1 endpoint method description

The decisive visual: what crosses the iteration boundary?



Evidence: mailboxes_and_arrows.html, beats 19–20; RQ1 endpoint method description

The decisive visual: what crosses the iteration boundary?



Evidence: mailboxes_and_arrows.html, beats 19–20; RQ1 endpoint method description

Iteration 2 revisits the same edges with different information

BP

$$Q_{2\to A} = L_2 + U_{B\to 2},$$
$$Q_{2\to B} = L_2 + U_{A\to 2}.$$

A receives $Q_{2\to A}$ B receives $Q_{2\to B}$

The two values can differ because they use different separately retained incoming U messages.

Projected GD

check A reads x_2^+, check B reads x_2^+.

A reads x_2^+ B reads x_2^+

Any effect that remains after the objective terms are summed, the step is selected and the box projection is applied is available only through the updated scalar.

The difference is not whether the graph is revisited. It is what information is available when it is revisited.

Evidence: RQ1-STATE-002; RQ1-ENDPOINT-001 validated native trace

Controlled test: remove only the recipient-specific distinction

$$Q^{\star}_{i\to a} = L_i + \sum_{b\in\mathcal{N}(i)\setminus\{a\}} U_{b\to i},$$

$$\bar{q}_i = \mathrm{mean}_{a\in\mathcal{N}(i)}\, Q^{\star}_{i\to a},$$

$$\delta_{i\to a} = Q^{\star}_{i\to a} - \bar{q}_i.$$

full sum-product

$$Q_{i\to a} = \bar{q}_i + \delta_{i\to a}$$

persistent-$\bar{q}$ ablation
exact check calculation

$$Q_{i\to a} = \bar{q}_i$$

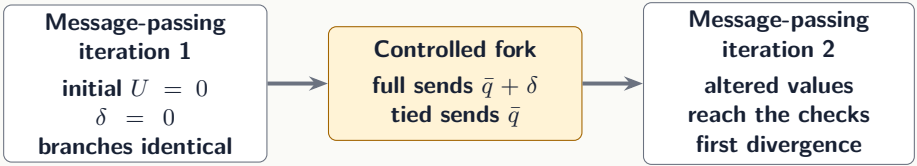
Held fixed

- parity-check matrix and Tanner graph;
- received samples and channel LLRs;
- exact check calculation;
- flooding schedule and initialization;
- stopping rule and 80-iteration limit;
- arrays remain indexed by Tanner-graph edge.

This is a controlled test inside exact sum-product BP. The ablation branch is not projected GD.

Evidence: RQ2-AWGN-001 experiment card, lines 24–64

Toy pulse: trajectories separate at the first possible point



Full branch at the shared bit

$$Q^{\star}_{2\to A} = \ln 3, \qquad Q^{\star}_{2\to B} = \ln(1/6).$$

hard-decision estimate 000 | syndrome 00

Tied branch at the shared bit

$$Q_{2\to A} = Q_{2\to B} = -\tfrac{1}{2}\ln 2.$$

hard-decision estimate 001 | syndrome 01

Reconstructing $\bar{q} + \delta$ restores the full trajectory; on the degree-one six-bit graph, $\delta = 0$ and tying is a null intervention.
This one-update snapshot may later rejoin full BP; it is not a convergence or performance result. The next slide is a separate persistent $N = 10200$ ablation.

Evidence: RQ1-STATE-002; mechanism report, lines 127–185

The recipient-specific distinction changes recovery on locked AWGN frames

Locked AWGN frames	Full	Persistent $\bar{q}$
Sealed descriptive anchor	24/24	0/24
Supporting 200-frame transfer	200/200	0/200

The counts are the same for both registered endpoints: parity validity and equality to the transmitted word.

Regular $(3, 6)$ LDPC, $N = 10200$, all-zero word, $E_s/N_0 = 1.5$ dB, 80-message-passing-iteration cap.

What the result supports

Removing $\delta_{i\to a}$ from the variable-to-check value changes recovery while the exact check calculation stays fixed.

200-frame qualification

The transfer used 4 Julia threads, 1 BLAS thread per process and gzip 1.14; the registered contract specifies 1 Julia thread, 1 BLAS thread and gzip 1.10.

The graph, flooding schedule, per-iteration check calculation and 80-iteration cap were fixed; removing $\delta_{i\to a}$ from Q changed recovery. Realized iteration count was an outcome.

Evidence: RQ2-AWGN-001 anchor and valid parallel transfer; mechanism report, lines 249–305

The update operator also changes the trajectory after one update

Direct BP map

$$\mathbf{u}^{+} = F_s(\mathbf{u})$$

The calculated check-to-variable state is assigned directly.

BP-residual gradient control

$$\mathbf{u}^{+} = \mathbf{u} - \alpha(I - J_s)^{\mathsf{T}}(\mathbf{u} - F_s(\mathbf{u})).$$

One gradient step is taken on $R_s(\mathbf{u}) = \frac{1}{2}\|\mathbf{u} - F_s(\mathbf{u})\|_2^2$,
where $J_s = \partial F_s(\mathbf{u})/\partial \mathbf{u}$.

	direct BP map	BP-residual gradient step
recipient-specific Q	(S, D)	(S, G)
tied $Q = \bar{q}$	(T, D)	(T, G)

After one update, the state, update and interaction vector contrasts are nonzero on the tree and cycle.

This 2×2 control uses BP edge coordinates and the squared objective $R_s(\mathbf{u}) = \frac{1}{2}\|\mathbf{u} - F_s(\mathbf{u})\|_2^2$—not the native projected-GD decoder. It shows that the recipient-specific Q calculation and next-state operator must be tested separately.

Evidence: RQ1-UPDATE-001 and RQ1-UPDATE-CYCLE-001

Current answer: what we know and what remains open

Established

- Within the controlled sum-product cases, Tanner-graph traversal alone is not a sufficient description.
- Recipient-specific variable-to-check information is causally important inside the tested recurrence.
- Discarding $\delta_{i \to a}$ from the variable-to-check value consumed by each check changed parity validity and equality to the transmitted word.
- A separate BP-coordinate control shows that the update operator also changes the trajectory after one update.

Still open

- The persistent-$\bar{q}$ sum-product ablation is not projected GD.
- $\bar{q}_i$ is not the native GD scalar x_i.
- The native methods also differ in state representation, parity calculation, objective, use of the channel LLR, projection, initialization, update and stopping.
- Recipient-specific variable-to-check information is not yet the unique explanation of the native performance gap.

Traversal alone is insufficient; recipient-specific variable-to-check information matters inside the tested exact sum-product recurrence; native BP–GD attribution remains open.

Evidence: mechanism report, outcome and RQ1 decision

Backup: both passes are $O(E)$, but not identical

Exact sum-product

- variable-to-check additions;
- nonlinear check calculation;
- separate Q/U messages on directed edges; U is retained for the next message-passing iteration;
- a-posteriori LLR and syndrome calculation.

Projected gradient descent

- products for check terms;
- partial derivatives of the objective function;
- $D_i = (2\mu_p \deg(i) + 8\mu_b)/n$, $d_i = -(\partial J/\partial x_i)/D_i$, then box projection;
- possible backtracking evaluations.

With bounded check degree, each graph evaluation is $O(E)$; bounded backtracking may repeat it. $O(E)$ does not imply equal constants, storage, work categories, evaluations or iterations.

Evidence: mechanism report, lines 313–322

Backup: evidence and claim provenance

Record	Role in this deck
`RQ1-STATE-002`	Three-bit tree and four-bit cycle: recipient-specific message intervention, reconstruction and null controls.
`RQ1-UPDATE-001`	Separate state and update contrasts on the tree.
`RQ1-UPDATE-CYCLE-001`	Structural confirmation of the state/update contrasts on a cycle.
`RQ1-ENDPOINT-001`	Validated native exact sum-product and projected gradient descent method descriptions.
`RQ2-AWGN-001 anchor`	Sealed 24-frame descriptive recovery consequence.
`RQ2-AWGN-001 transfer`	Valid supporting 200-frame transfer; not the canonical registered one-thread confirmation.

Primary narrative source: `mechanism_report.md`. Experiment cards and retained manifests define the exact intervention, implementations, frozen protocol and evidence boundary.

Repository state used for the deck: main at `ad4d6e4`; quantitative evidence retained under `experiments/rq1_*` and `experiments/rq2_qbar_awgn_001`.